

A small icon of a flag with the number '9' on it, indicating the slide number.

Processor Level Design



1. Processor Architecture and Performance
2. Processor Performance Measurement
- 3. Queueing Models**

- the rate at which tasks requiring service arrive = λ
- the rate at which the tasks are serviced = μ

Mean Utilization

$$\rho = \frac{\lambda}{\mu}$$

ρ = mean utilization

The average number of tasks queue in the system, including tasks waiting for service and those actually being served,

$$\ell_Q = \frac{\rho}{1 - \rho}$$

The average number of tasks waiting in the queue, excluding those being served,

$$\ell_w = \frac{\lambda^2}{\mu(\mu - \lambda)} = \ell_q - \rho$$

The average time that arriving task spent in the system, both waiting for service and being served,

$$t_q = \frac{\ell_q}{\lambda} = \frac{1}{\mu - \lambda}$$

The mean time spent waiting in the queue, excluding service time,

$$t_w = \frac{\lambda}{\mu(\mu - \lambda)}$$

Problem 1

Suppose that CPU behavior in a multiprogramming system can be analyzed using the M/M/1 queueing model. Programs are sent to the CPU for execution at a mean rate of **eight programs per minute** and are executed on a first-come first-served basis. The average program requires **six seconds** of CPU execution time,

- (a) What is the mean time between program arrivals at the CPU?
- (b) What is the mean number of programs waiting for CPU execution to be completed?
- (c) What is the mean time a program must wait for its execution to be completed?

solution

$$\lambda = 8 \text{ programs/ min} = 8/60 \text{ programs/ sec}$$

$$6 \text{ sec} = 1 \text{ program}$$

$$1 \text{ sec} = 1/6 \text{ program}$$

$$\mu = 1/6 \text{ program/ sec}$$

Queueing Models



(a) $8 \text{ programs} = 60 \text{ sec}$

$1 \text{ program} = ?$

$$= 60 \times 1/8 = 7.5 \text{ sec}$$

(b) $\ell_Q = ?$ $\ell_Q = \frac{\rho}{1 - \rho}$

$$\rho = \frac{\lambda}{\mu} = \frac{8}{60} \times \frac{6}{1} = 0.8$$

$$\ell_Q = \frac{\rho}{1 - \rho} = \frac{0.8}{1 - 0.8} = 4 \text{ programs}$$

(c) $t_Q = ?$

$$t_Q = \frac{\ell_Q}{\lambda}$$

$$t_Q = \frac{4}{8/60} = 30 \text{ sec}$$

Problem 2

Suppose that people arrive at a public telephone booth at an average rate of **10 per hour**. The lengths of the calls made from the booth are found to have a negative exponential distribution with a mean length of **2.5 minutes**,

- (a) What is the probability that someone arriving at the telephone booth will find it occupied?
- (b) The telephone company will install a second booth if a customer must wait an average of four minutes or more to gain access to the first telephone. By how much must the flow of customers to the first telephone increase in order for the telephone company to install the second phone?

solution

$$\lambda = 10 \text{ persons/ hour} = 10/60 \text{ persons/ min}$$

$$2.5 \text{ min} = 1 \text{ person}$$

$$1 \text{ min} = 1/2.5 \text{ person/ min}$$

$$\text{mean service rate, } \mu = 1/2.5 \text{ person/ min}$$

(a) busy, $\rho = ?$

$$\rho = \frac{\lambda}{\mu} = 10/60 \times 2.5 = 0.42$$

(b)

$$t_w \geq 4 \text{ min}$$

Increase amount of person = ?

$$\begin{aligned} t_w &= \frac{\lambda'}{\mu(\mu - \lambda')} \geq 4 \\ &= \frac{\lambda'}{\frac{1}{2.5}(\frac{1}{2.5} - \lambda')} \geq 4 \end{aligned}$$

$$\lambda' \geq 0.246 \text{ person/min}$$

$$1 \text{ min} = 0.246 \text{ person}$$

$$1 \text{ hour} = ?$$

$$= 0.246 \times 60 = 14.76 \text{ person}$$

$$\text{Increase amount of person} = \lambda' - \lambda$$

$$= 14.76 - 10$$

$$= 4.76 \text{ person per hour}$$

Problem 3

A certain computer system executes a stream of tasks in a manner that can be accurately modeled by an M/M/1 queueing system. The computer is **busy 75 percent** of the time, and the average job spends **four minutes** in the computer,

- (a) How many jobs are in the computer on average?
- (b) What is the maximum rate at which jobs may arrive at the system before it becomes overloaded? State clearly your definition of overloaded.

solution

$$\text{busy, } \rho = 0.75 \qquad t_Q = 4 \text{ min}$$

$$(a) \quad l_Q = \frac{\rho}{1 - \rho} = \frac{0.75}{1 - 0.75} = 3 \text{ jobs}$$

(b) $\mu = ?$

$$t_Q = \frac{l_Q}{\lambda} = \frac{l_Q}{\rho \mu}$$

$$\begin{aligned}\mu &= \frac{l_Q}{\rho \times t_Q} \\ &= \frac{3 \text{ jobs}}{4 \text{ min} \times 0.75} \\ &= 1 \text{ job / min}\end{aligned}$$

The system becomes overload if $\lambda \geq 1$.



THANK YOU FOR YOUR ATTENTION